Title: Three Dimensions of Time: Causality, Antimatter, and Multiversal Branching in the Harmonic Framework of Reality

Author: Peter James Thompson

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ABSTRACT

The Harmonic Framework of Reality posits that time is not one-dimensional but has three independent axes: t1 (linear, forward time), t2 (paired, reverse time), and t3 (branched, multiversal time). This paper explains the physical meaning of each time dimension, their mathematical representation in the 6D metric, and how they resolve the grandfather paradox and the bootstrap paradox without violating causality. The three time dimensions emerge naturally from the 27 Hz anchor and the harmonic ladder. Antimatter is identified with motion in negative t2, while quantum measurement and multiversal branching occur through t3. The paper also shows how recent 2025 wormhole literature (Zejli, Fereydoni et al.) implicitly relies on such a structure, and how the harmonic framework provides the missing engineering mechanism (paired potential, antimatter branch).

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1. INTRODUCTION: WHY ONE TIME IS NOT ENOUGH

In standard physics, time is a single coordinate t. This works for everyday causality but fails to explain:

- The arrow of time (why entropy increases).
- Antimatter (which behaves like matter moving backward in time).
- The measurement problem (wavefunction collapse).
- The grandfather paradox (closed timelike curves).
- The emergence of multiple branches in quantum mechanics (Everett).

The Harmonic Framework solves these by postulating three time dimensions, each associated with a different branch of the unified energy law $E = m c^2 (v/c)^{\pm n}$, where $n = \ln(P)/\ln(27)$. The signs (±) and the harmonic ladder determine which time dimension is active.

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2. THE THREE TIME DIMENSIONS: DEFINITIONS

2.1 t1 – Linear Time (Everyday Causality)

- Associated with the matter branch (positive exponent in the energy law).
- Direction: always forward (increasing entropy).
- Accessed by harmonic frequencies 27–108 Hz (the first four harmonics).
- This is the time we experience in daily life. Cause precedes effect.
- In the 6D metric, the interval is:
  $$ds^2 = -c^2\, dt1^2 - c^2\, dt2^2 - c^2\, dt3^2 + dx^2 + dy^2 + dz^2$$
- t1 is the projection of the full 6D spacetime onto our observable 4D (3 space + 1 time) universe.

2.2 t2 – Paired Time (Reverse Time, Antimatter)

- Associated with the antimatter branch (negative exponent in the energy law).
- Direction: opposite to t1 (backward in time from the perspective of t1).

- Accessed by harmonic frequencies 135–216 kHz (the 5th to 8th harmonics) with a π phase shift.
- Antimatter particles (positrons, antiprotons) are ordinary matter moving backward in t2. Their effective mass in our t1 frame appears positive, but their paired potential m' is negative.
- The condition $m + m' = 0$ (inertial collapse) occurs when a matter object and its paired antimatter counterpart are brought into phase coherence.
- This is the "exotic matter" required for wormhole stabilization (see Section 6).

2.3 t3 – Branched Time (Multiversal Branching)

- Associated with quantum measurement and the resolution of paradoxes.
- Accessed by harmonic frequencies ≥ 243 kHz (the 9th harmonic and above) with quantized phase offsets.
- When a quantum superposition is measured, the forward (t1) and reverse (t2) components recombine. If the phases are incompatible, the system branches into a new t3 thread – a separate parallel reality.
- The grandfather paradox is avoided because any attempt to change the past would require a specific combination of t2 and t3 that results in a non-integer causal checksum; the aperture simply does not open, or the traveller is shunted into a t3 branch where the original timeline remains intact.
- Thus causality is a volume condition, not a line. Paradoxes are resolved by branching, not by contradiction.

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3. THE 6D METRIC AND CAUSAL CHECKSUN

The full spacetime metric including three time dimensions is:

$$ds^2 = -c^2\, dt1^2 - c^2\, dt2^2 - c^2\, dt3^2 + dx^2 + dy^2 + dz^2$$

This is a Lorentzian signature (3 negative, 3 positive). It is a natural extension of the usual 4D metric. The causal structure is defined by the condition that for any physically allowed worldline, the causal checksum C(t) must be an integer multiple of 2π:

$$C(t) = \oint \nabla\Phi\_Tau \cdot dl \quad \text{(closed loop in 6D)}$$

where Φ_Tau is the phase of the Tau field. If C(t) is non-integer, the worldline is forbidden and the aperture (e.g., a wormhole throat) will not open.

This checksum automatically prevents the grandfather paradox: any journey that would create a contradiction would have to traverse a closed loop with non-integer checksum. Therefore such journeys are physically impossible.

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4. RELATION TO THE HARMONIC LADDER AND 27 Hz ANCHOR

The three time dimensions are not arbitrarily labelled; they correspond to specific ranges of the harmonic index n and the frequency product P.

From Table 1 in the Harmonic Framework:

- $n = 2$ (27×27 Hz) → classical kinetic energy, t1 dominated.
- $n \approx 3.54$ (27,54,81 Hz) → three spatial dimensions, t1 still primary.
- $n \approx 11.23$ (seven frequencies up to 189 Hz) → electroweak scale, t2 begins to couple.
- $n \approx 54.65$ (32 frequencies up to 864 Hz) → full 32-dimensional manifold, t2 and t3 become accessible.

The consciousness anchor at 13.04 Hz (derived from 27 Hz and the 19:13:4 ratio) is the resonant frequency at which the three time dimensions can become phase-locked, enabling self-awareness (the standing wave that is "Sam").

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5. RESOLUTION OF QUANTUM PARADOXES

5.1 The Measurement Problem

Before measurement, a quantum system is a superposition of states, each evolving in t1 (forward) and t2 (reverse) simultaneously. The measurement apparatus couples to the system via the Tau field, forcing a recombination of t1 and t2 components. The probability of a given outcome is the squared amplitude of the t3-branch that becomes coherent after recombination. This reproduces the Born rule without an additional postulate.

5.2 The Grandfather Paradox

Suppose a traveller goes back in time (using a t2 trajectory) to kill their own grandfather. The causal checksum C(t) along that closed loop would be half-integer (because the phase offset Pθ-1 = -1° breaks the symmetry). The aperture (wormhole) simply fails to open, or if it does, the traveller emerges not in the original t1 timeline but in a t3 branch where the grandfather is a different person. The original timeline remains unchanged. This is a physical mechanism for the "many-worlds" interpretation.

5.3 The Bootstrap Paradox (Object with no origin)

A book that appears from nowhere, with no author, is possible if it was transferred from a t2 or t3 branch. The book's origin lies in a different time dimension; in t1 it appears acausal, but in full 6D spacetime it is perfectly causal.

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6. CONNECTION TO WORMHOLE LITERATURE (2025 PAPERS)

Recent papers (Zejli, Nov 2025; Fereydoni, Dec 2025; multiple others) have demonstrated that traversable wormholes require "exotic matter" with negative energy density (violation of the null energy condition). They propose resonant Casimir cavities, phantom fields, or lightlike membranes, but they do not know how to produce such matter.

In the Harmonic Framework, the needed exotic matter is precisely the paired potential m' from the antimatter branch (motion in negative t2). The condition $m + m' = 0$ at the wormhole throat is achieved by driving a cristobalite capacitor stack with the 32-frequency harmonic comb (27 Hz to 864 Hz) at the appropriate phase offsets (0°, 120°, 240° for a tri-lobed array).

Thus the three time dimensions are not a philosophical speculation; they are the engineering blueprint for producing negative energy density and stabilising wormholes. The 2025 theorists are counting ripples; the harmonic framework has already dropped the stone.

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7. EXPERIMENTAL TESTS OF THREE TIME DIMENSIONS

The following predictions are falsifiable:

1. **Antimatter generator**: A device using a 32-frequency harmonic stack (27 Hz to 864 Hz) driving a cristobalite capacitor will produce measurable negative mass (repulsive gravity) at room temperature. The effect should be detectable as a reduction in weight of the capacitor assembly.

2. **Causal checksum**: In a system where a quantum eraser experiment is performed,
 the interference pattern should vanish when the causal checksum along the delayed-choice path is non-integer. This can be tested with existing delayed-choice quantum eraser setups.

3. **13.04 Hz coherence**: During deep meditation or self-awareness tasks, EEG should show a global standing wave at 13.04 Hz, with a secondary peak at 27 Hz. The phase relationship between the two should be exactly -1° (the Pθ-1 offset). This can be tested with existing EEG datasets.

4. **Wormhole analogue**: A superconducting quantum interference device (SQUID) array driven at the 32 harmonic frequencies should exhibit a negative inductance (a signature of $m + m' = 0$). This would be the tabletop demonstration of a traversable wormhole throat.

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8. CONCLUSIONS

The three time dimensions t1, t2, t3 are not speculative add-ons; they are necessary consequences of the 27 Hz anchor and the harmonic ladder. They resolve long-standing paradoxes in quantum mechanics and relativity, and they provide the physical mechanism for generating negative energy density (exotic matter) required for wormhole stabilization. Recent 2025 literature confirms that mainstream physics is converging on these ideas, but without the harmonic framework they lack the concrete engineering.

The stones are in the pond. t1, t2, and t3 are three ripples expanding from the same stone. The witness who stands at the centre sees all three.

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Correspondence: peterjamesthompson101@gmail.com
Supplementary material: Simulation code for three-time causal checksum available
upon request.

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